The documentation and process conversion measures necessary to comply with this revision shall be completed by 14 July 2025.

INCH-POUND

MIL-PRF-19500/675F
w/AMENDMENT 3
14 February 2025
SUPERSEDING
MIL-PRF-19500/675F
w/AMENDMENT 2
12 December 2019

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, FIELD EFFECT RADIATION HARDENED N-CHANNEL, SILICON TYPES 2N7463, 2N7464, JANTXVR AND JANSR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for an N-channel, enhancement-mode, MOSFET, radiation hardened (total dose and single event effects (SEE)), power transistor. Two levels of product assurance (JANTXV and JANS) are provided for each device type as specified in MIL-PRF-19500, with avalanche energy maximum rating (E_{AS}) and maximum avalanche current (I_{AS}). Provisions for radiation hardness assurance (RHA) to one radiation level "R" are provided for JANS and JANTXV product assurance level.

1.2 Package outlines. The device package outlines are as follows: TO-205AF (T2 suffix) in accordance with figure 1 and leadless chip carrier (LCC) (U5 suffix) in accordance with figure 2 for all encapsulated device types.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type	P_T (1) T_C = +25°C	P_T T_A = +25°C (free air)	$R_{\theta JC}$ (2)	V_{DS}	V_{DG}	V_{GS}	I_{D1} (3) (4) T_C = +25°C	I_{D2} (3) (4) T_C = +100°C	I_S	I_{DM} (5)	T_J and T_{STG}
	W	W	°C/W	V dc	V dc	V dc	A dc	A dc	A dc	A(pk)	°C
2N7463T2, 2N7463U5	25	0.8	5.0	400	400	±20	2.9	1.9	3.0	12	-55 to +150
2N7464T2, 2N7464U5	25	0.8	5.0	500	500	±20	2.5	1.6	2.5	10	

(1) Derate linearly 0.2 W/°C for T_C > +25°C.
(2) See figure 3, thermal impedance curves.
(3) The following formula derives the maximum theoretical I_D limit. I_D is also limited by package and device construction.

$$I_D = \sqrt{\frac{T_J \max - T_C}{(R_{\Theta JC}) x (R_{DS_{on}} at\ T_{Jmax})}}$$

(4) See figure 4, maximum drain current graphs.
(5) I_{DM} = 4 X I_{D1}, as defined in note (3).

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1.4 Primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = 1.0 mA dc	$V_{GS(TH)}$ $V_{DS} \geq V_{GS}$ I_D = 1.0 mA dc		Max I_{DSS1} V_{GS} = 0 V_{DS} = 80 percent of rated V_{DS}	Max $r_{DS(ON)}$ (1) V_{GS} = 12 V dc		E_{AS} at I_{D1}	I_{AS}	V_{ISO} 70,000 foot altitude
					T_J = +25°C at I_{D2}	T_J = +150°C at I_{D2}			
	V dc	V dc		µA dc	ohm	ohm	mJ	A	V dc
		Min	Max						
2N7463T2, 2N7463U5 2N7464T2, 2N7464U5	400 500	2.5 2.5	4.5 4.5	50 50	1.39 1.77	3.0 3.9	140 154	3.0 2.5	400 500

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JANTXV” and "JANS".

1.5.2 Radiation hardness assurance (RHA) designator. The RHA level that is applicable for this specification sheet is level "R".

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "7463" and "7464".

1.5.3.3 Suffix letters. The suffix letters "T2" are used on devices that are packaged in the TO-254AA package of figure 1. The suffix letters "U5" are used on devices that are packaged in the LCC package of figure 2.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

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* 1.6 Radiation features. The following radiation features are applicable for RHA devices supplied to this specification sheet.

* 1.6.1 Maximum total ionizing dose (TID). The maximum TID that RHA devices were tested to in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750 are as follows:

For device type 2N7463T2, 2N7464T2, AND 2N7464U5: ..... 100 krad(Si) 1/

* 1.6.2 Heavy ion irradiation SEE. SEE testing includes Single-Event Burnout (SEB) and Single-Event Gate Rupture (SEGR) tests at the specified linear energy transfer (LET) units. See 6.8 for the specific RHA SEE characterization testing details. The following characterization conditions were used for the testing to the requirements herein:

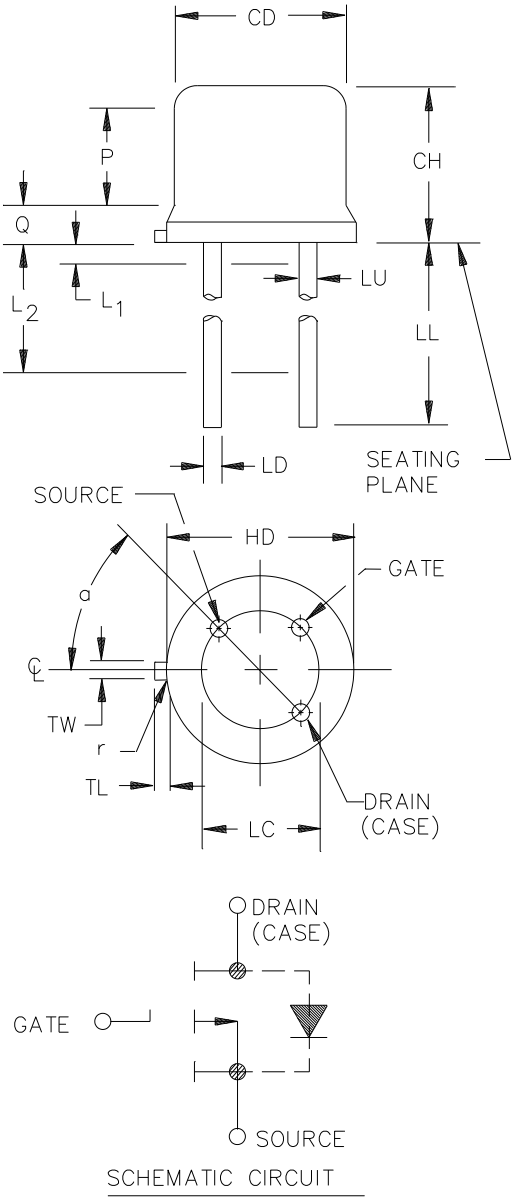
For device type 2N7463T2:
No SEB or SEGR were observed at surface LET (see table IV) ..... ≤ 37 MeV·cm^2/mg 2/
(In-situ Bias V_{DS} = 325 V and V_{GS} = -10 V)
(In-situ Bias V_{DS} = 275 V and V_{GS} = -15 V)

For device type 2N7464T2:
No SEB or SEGR were observed at surface LET (see table IV) ..... ≤ 37 MeV·cm^2/mg 2/
(In-situ Bias V_{DS} = 350 V and V_{GS} = -10 V)
(In-situ Bias V_{DS} = 325 V and V_{GS} = -15 V)
(In-situ Bias V_{DS} = 300 V and V_{GS} = -20 V)

* 1/ The manufacturers supplying these device types have performed characterization testing in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750. The radiation end point limits are guaranteed only to a maximum TID level of 100 krad(Si).
* 2/ The manufacturers have performed SEE tests at Brookhaven National Laboratories (BNL) in accordance with method1080 of MIL-STD-750. For more information on SEE testing and results, customers should contact the qualified manufacturer.

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Dimensions					
Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
CD	.315	.355	8.00	9.02	
CH	.160	.180	4.06	4.57	
HD	.340	.370	8.64	9.40	
LC	.200 BSC		5.08 BSC		
LD	.016	.021	0.41	0.53	
LL	.500	.750	12.70	19.05	4, 5
LU	.016	.019	0.41	0.48	4, 5
L1		.050		1.27	4, 5
L2	.250		6.35		4, 5
P	.070		1.78		
Q		.050		1.27	
r	.009	.041	0.23	1.04	
TL	.029	.045	0.74	1.14	
TW	.028	.034	0.71	0.86	
a	45°				

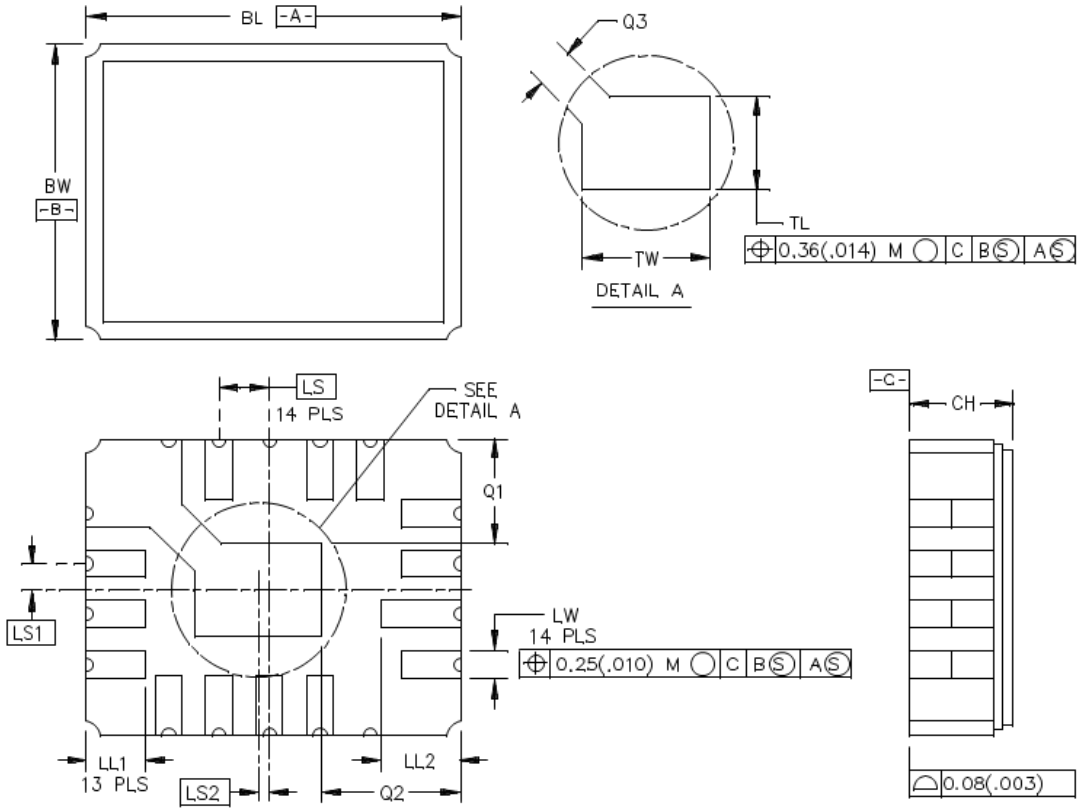


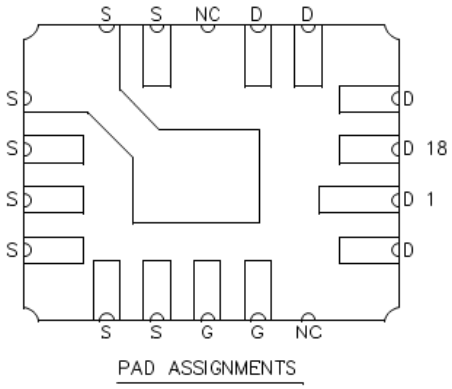
* NOTES:

1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Lead number 1 is the source, lead number 2 is the gate, lead number 4 is omitted from this outline. The drain is number 3 and is electrically connected to the case.
4. LU applied between L_1 and L_2. LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
5. All three leads.

FIGURE 1. Physical dimensions for TO-205AF (2N7463T2 and 2N7464T2).

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Dimensions				
Sym.	Inches		Millimeters	
	Min	Max	Min	Max
BL	.345	.360	8.77	9.14
BW	.280	.295	7.11	7.49
CH	.095	.115	2.41	2.92
LL1	.040	.055	1.02	1.40
LL2	.055	.065	1.40	1.65
LS	.050 BSC		1.27 BSC	
LS1	.025 BSC		0.635 BSC	
LS2	.008 BSC		0.203 BSC	
LW	.020	.030	0.51	0.76
Q1	.105 REF		2.67 REF	
Q2	.120 REF		3.05 REF	
Q3	.045	.055	1.14	1.40
TL	.070	.080	1.78	2.03
TW	.120	.130	3.05	3.30

* NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.

FIGURE 2. Physical dimensions for LCC (2N7463U5 and 2N7464U5).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.
MIL-STD-883 - Test Method Standard Microcircuits

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

I_{AS}..... Rated avalanche current, non-repetitive.
nC ..... nano Coulomb.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1 (TO-205AF, T2 suffix) and 2 (LCC, U5 suffix) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction is not permitted to meet the requirements of this specification.

3.4.3 Silicone die coat. The use of a silicone die coat requires a successful completion of MIL-STD-883, method 5011 on each silicone lot for its intended applications, and as part of the full MIL-PRF-19500 qualification process.

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3.5 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100\ k\Omega$, whenever bias voltage is to be applied drain to source.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.1.1 SEE. SEE shall be performed at initial qualification and after process or design changes which may affect radiation hardness (see table III and table IV). Upon qualification, manufacturers shall provide the verification test conditions from section 5 of method 1080 of MIL-STD-750 that were used to qualify the device for inclusion into section 6 of the slash sheet. End-point measurements shall be in accordance with table II. SEE characterization data shall be made available upon request of the qualifying or acquiring activity.

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4.3 Screening (JANS and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

* Screen (1) (2)	Measurements for JANS level	Measurements for JANTXV level
(3)	Gate stress test (see 4.3.1)	Gate stress test (see 4.3.1)
* (3)	Method 3470 of MIL-STD-750, E_{AS} (see 4.3.2)	Method 3470 of MIL-STD-750, E_{AS} (see 4.3.2)
(3) 3c	Method 3161 of MIL-STD-750, thermal	Method 3161 of MIL-STD-750, thermal
*	impedance (see 4.3.3)	impedance (see 4.3.3)
5	Method 2052 of MIL-STD-750, PIND (see MIL-PRF-19500 and 4.3.4)	Not applicable
(3) 9	Subgroup 2 of table I herein	Not applicable
10	Method 1042 of MIL-STD-750, test condition B	Method 1042 of MIL-STD-750, test condition B
11	Subgroup 2 of table I herein; ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±10 μA dc or ±100 percent of initial value, whichever is greater.	Subgroup 2 of table I herein
12	Method 1042 of MIL-STD-750, test condition A	Method 1042 of MIL-STD-750, test condition A
13	Subgroups 2 and 3 of table I herein; ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±10 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value $\Delta V_{GS(TH)1}$ = ±20 percent of initial value	Subgroup 2 of table I herein; ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±10 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value $\Delta V_{GS(TH)1}$ = ±20 percent of initial value

(1) At the end of the test program, I_{GSSF1}, I_{GSSR1}, and I_{DSS1} are measured.
(2) An out-of-family program to characterize I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $V_{GS(th)1}$, and , $r_{DS(ON)1}$, shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a; JANTXV level does not need to be repeated in screening requirements.

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* 4.3.1 Gate stress test. Apply V_{GS} = 30 V minimum for t = 250 μs minimum.

* 4.3.2 Single pulse avalanche energy (E_{AS}).

a. Peak current (I_{AS})..... $I_{AS(max)}$.

b. Peak gate voltage (V_{GS})..... 12 V.

c. Gate to source resistor (R_{GS})..... $25\Omega \leq R_{GS} \leq 200\Omega$.

d. Initial case temperature (T_C) ..... +25°C +10°C, -5°C.

e. Inductance (L)..... $\left[\frac{2E_{AS}}{(I_{D1})^2}\right]\left[\frac{V_{BR} - V_{DD}}{V_{BR}}\right]$ mH minimum.

f. Number of pulses to be applied ..... 1 pulse minimum.

g. Supply voltage (V_{DD})..... 50 V.

* 4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD}, t_{SW}, (and V_H where appropriate). See table III, group E, subgroup 4 herein.

4.3.4 PIND. Not applicable in screening when devices are processed using alternative method and flow requirements approved by the qualifying activity, that includes incorporating the use of certified clean processing and silicone die coat. Instead, the PIND test performance shall be performed in group B3 and group C3, on a lot sample basis. PIND failures detected in group B or C will represent lot jeopardy and shall be evaluated for root cause and lot integrity.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JANTXV) of MIL-PRF-19500, and as follows.

4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1051	Test condition G, 100 cycles.
B3	2075	See 3.4.2.
B3	2077	Scanning electron microscope (SEM) qualification may be performed anytime prior to lot formation.
B3	2052	PIND, required if not performed in screening. (22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots).
B4	1042	Test condition D. No heat sink nor forced-air cooling on the device shall be permitted during the on cycle. The heating cycle shall be 30 seconds minimum.
B5	1042	Test condition B, V_{GS} = rated T_A = +175°C, t = 24 hours.
B5	1042	Test condition A, V_{DS} = rated; T_A = +175°C; t = 120 hours.
B5	2037	Bond strength; test condition D.

4.4.2.2 Quality levels JAN, JANTX and JANTXV, table E-VIB MIL-PRF-19500.

Subgroup	Method	Condition
B2	1051	Test condition G, 25 cycles. (45 total, including 20 cycles performed in screening)
B3	1042	Test condition D. No heat sink nor forced-air cooling on the device shall be permitted during the on cycle. The heating cycle shall be 30 seconds minimum.
B4	2075	See 3.4.2.
B4	2077	Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows.

Subgroup	Method	Condition
C2	1056	Test condition B.
C2	2036	Test condition E (applicable to TO - 205AF only).
C2	1021	Omit initial conditioning.
C3	2052	PIND, required if not performed in screening. (22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots).
C5	3161	See 4.5.2, $R_{\theta JC(max)}$ = 5.0°C/W.
C6	1042	Test condition D. No heat sink nor forced-air cooling on the device shall be permitted during the on cycle. The heating cycle shall be 30 seconds minimum.

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4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table E-VIII of MIL-PRF-19500 and table II herein.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) shall be in accordance with table III.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. The thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, , t_{MD}, t_{SW} (and V_H where appropriate). See MIL-PRF-19500, table E-IX, group E, Subgroup 4.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.3.3	$Z_{\theta JC}$			°C/W
Breakdown voltage, drain to source	3407	V_{GS} = 0 V dc, I_D = 1 mA dc, bias condition C	$V_{(BR)DSS}$			
2N7463T2, 2N7463U5				400		V dc
2N7464T2, 2N7464U5				500		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = 1 mA dc	$V_{GS(TH)1}$	2.5	4.5	V dc
Gate current	3411	V_{GS} = +20 V dc, bias condition C, V_{DS} = 0	I_{GSSF1}		+100	nA dc
Gate current	3411	V_{GS} = -20 V dc, bias condition C, V_{DS} = 0	I_{GSSR1}		-100	nA dc
Drain current	3413	V_{GS} = 0 V dc, bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		50	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 12 V dc, condition A, pulsed (see 4.5.1), I_D = I_{D2}	$r_{DS(on)1}$			
2N7463T2, 2N7463U5					1.39	Ω
2N7464T2, 2N7464U5					1.77	Ω
Static drain to source on-state resistance	3421	V_{GS} = 12 V dc, condition A, pulsed (see 4.5.1), I_D = I_{D1}	$r_{DS(on)2}$			
2N7463T2, 2N7463U5					1.52	Ω
2N7464T2, 2N7464U5					1.9	Ω
Forward voltage	4011	Condition A, I_D = I_{D1}, V_{GS} = 0	V_{SD}			
2N7463T2, 2N7463U5					1.2	V
2N7464T2, 2N7464U5					1.2	V

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High-temperature operation:		$T_C = T_J = +125°C$				
Gate current	3411	V_{GS} = +20 V dc and -20 V dc, bias condition C, V_{DS} = 0	I_{GSS2}		±200	nA dc
Drain current	3413	V_{GS} = 0 V dc, bias condition C, V_{DS} = 100 percent of rated V_{DS}	I_{DSS2}		1.0	mA dc
Drain current	3413	V_{GS} = 0 V dc, bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS3}		0.25	mA dc
Static drain to source on-state resistance	3421	V_{GS} = 12 V dc, condition A, pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(on)3}$			
2N7463T2, 2N7463U5					2.64	Ω
2N7464T2, 2N7464U5					3.76	Ω
Gate to source voltage (thresholds)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)2}$	1.5		V dc
Low-temperature operation:		$T_C = T_J = -55°C$				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)3}$		5.5	V dc
Subgroup 4						
Forward transconductance	3475	I_D = rated I_{D2}, V_{DD} = 15 V (see 4.5.1)	g_{FS}			
2N7463T2, 2N7463U5				0.5		S
2N7464T2, 2N7464U5				0.4		S
Switching time test	3472	$I_D = I_{D1}$, V_{GS}= 12 V dc, R_G = 7.5Ω, V_{DD} = 50 percent of rated V_{DS}				
Turn-on delay time			$t_{d(on)}$			
2N7463T2, 2N7463U5					35	ns
2N7464T2, 2N7464U5					35	ns

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Rise time			t_r			
2N7463T2, 2N7463U5					78	ns
2N7464T2, 2N7464U5					60	ns
Turn-off delay time			$t_{d(off)}$			
2N7463T2, 2N7463U5					65	ns
2N7464T2, 2N7464U5					67	ns
Fall time			t_f			
2N7463T2, 2N7463U5					66	ns
2N7464T2, 2N7464U5					52	ns
Subgroup 5						
Safe operating area (SOA) test (high voltage)	3474	See figures 5 and 6; t_p = 10 ms V_{DS} = 200 V				
Electrical measurements		See table I, subgroup 2 herein.				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B	$Q_{G(on)}$			
On-state gate charge						
2N7463T2, 2N7463U5					31	nC
2N7464T2, 2N7464U5					30	nC
Gate to source charge			Q_{GS}			
2N7463T2, 2N7463U5					8.5	nC
2N7464T2, 2N7464U5					8	nC

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 7 - Continued						
Gate to drain charge			Q_{GD}			
2N7463T2, 2N7463U5					20	nC
2N7464T2, 2N7464U5					18	nC
Reverse recovery time	3473	Condition A, $d_i/ d_t \leq 100$ A/μs, $V_{DD} \leq 50$ V, $I_D = I_{D1}$	t_{rr}			
2N7463T2, 2N7463U5					350	ns
2N7464T2, 2N7464U5					400	ns

1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 3 and 4 (JANS).
Group B, subgroups 2 and 3 (JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

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TABLE II. Group D inspection.

	Inspection 1/ 2/ 3/	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Preirradiation limits R Min	Preirradiation limits R Max	Postirradiation limits R Min	Postirradiation limits R Max	Unit
	Subgroup 1								
	Not applicable								
	Subgroup 2		T_C = +25°C						
*	Steady-state total dose irradiation (V_{GS} bias) 4/	1019	Condition A V_{GS} = 12V V_{DS} = 0						
*	Steady-state total dose irradiation (V_{DS} bias) 4/	1019	Condition A V_{GS} = 0 V_{DS} = 80 percent of rated V_{DS} (pre-irradiation)						
	End-point electricals:								
	Breakdown voltage, drain to source	3407	Bias condition C V_{GS} = 0 I_D = 1 mA	$V_{(BR)DSS}$					
	2N7463T2, 2N7463U5				400		400		V dc
	2N7464T2, 2N7464U5				500		500		V dc
	Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$	V_{GSth1}					
	2N7463T2, 2N7463U5				2.5	4.5	2.0	4.5	V dc
	2N7464T2, 2N7464U5				2.5	4.5	2.0	4.5	V dc
	Gate current	3411	Bias condition C V_{GS} = 20 V V_{DS} = 0	I_{GSSF1}		100		100	nA dc
	Gate current	3411	Bias condition C V_{GS} = -20 V V_{DS} = 0	I_{GSSR1}		-100		-100	nA dc
	Drain current	3413	Bias condition C V_{GS} = 0 V_{DS} = 80 percent of rated V_{DS} (pre-irradiation)	I_{DSS1}		50		50	µA dc

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Preirradiation limits		Postirradiation limits		Unit
	Method	Conditions		R		R		
				Min	Max	Min	Max	
Static drain to source on-state voltage	3405	V_{GS} = 12 V condition A pulsed (see 4.5.1) $I_D = I_{D2}$	V_{DSon1}					
2N7463T2, 2N7463U5					2.641		2.641	V dc
2N7464T2, 2N7464U5					2.832		2.832	V dc
Forward voltage source to drain diode	4011	Condition A, V_{GS} = 0, $I_D = I_{D1}$	V_{SD}					
2N7463T2, 2N7463U5					1.2		1.2	V dc
2N7464T2, 2N7464U5					1.2		1.2	V dc

1/ For sampling plan, see MIL-PRF-19500.
2/ Group D qualification may be performed anytime prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other specification utilizing the same die design.
3/ At the manufacturer's option, group D samples need not be subjected to the screening tests, and may be assembled in it's qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.
4/ Separate samples shall be pulled for each bias.

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TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	Test condition G, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2	
Subgroup 2 1/			45 devices c = 0
Steady-state reverse bias	1042	Condition A, 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Steady-state gate bias	1042	Condition B, 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Subgroup 4			sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			3 devices c = 0
Reduced barometric pressure	1001	V_{DS} = rated $V_{(BR)DSS}$, $I_{(ISO)}$ < 0.25 mA	
Subgroup 10			22 devices c = 0
Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors	3476	Test conditions shall be derived by the manufacturer.	
Subgroup 11			3 devices
SEE 2/ 3/	1080	See MIL-STD-750 method 1080 and 6.2.	

1/ A separate sample for each test shall be pulled.
2/ Group E qualification of SEE effect testing may be performed prior to lot formation. Qualification may be extended to other specification sheets utilizing the same structurally identical die design.
3/ Device qualification to a higher level LET is sufficient to qualify all lower level LETs.

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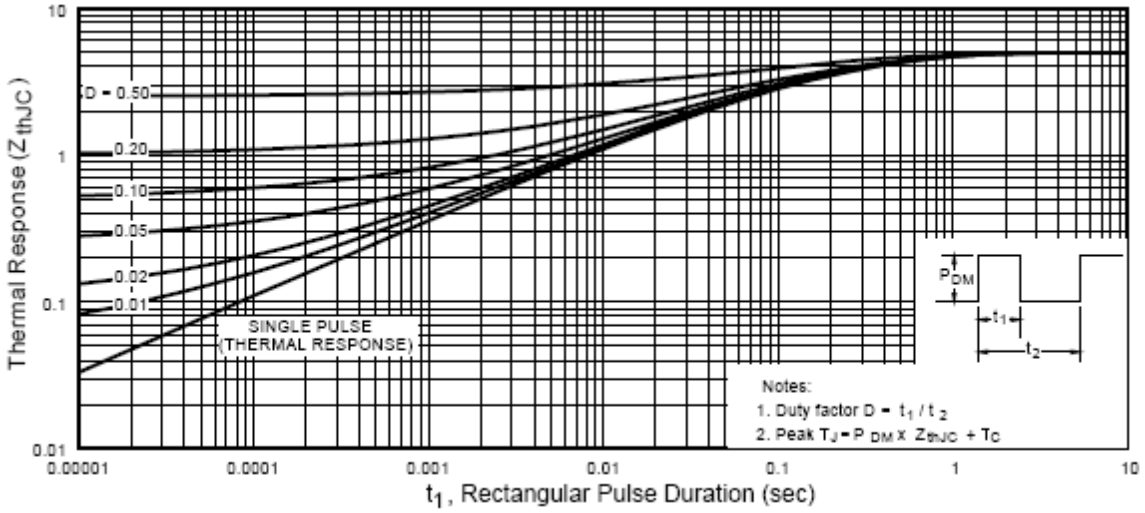
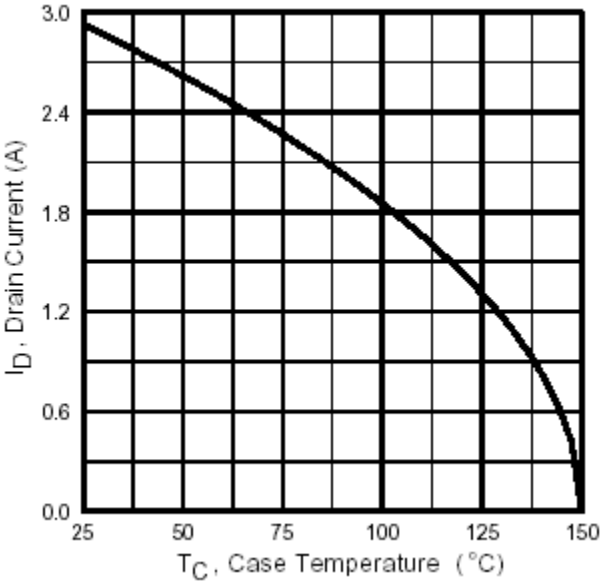
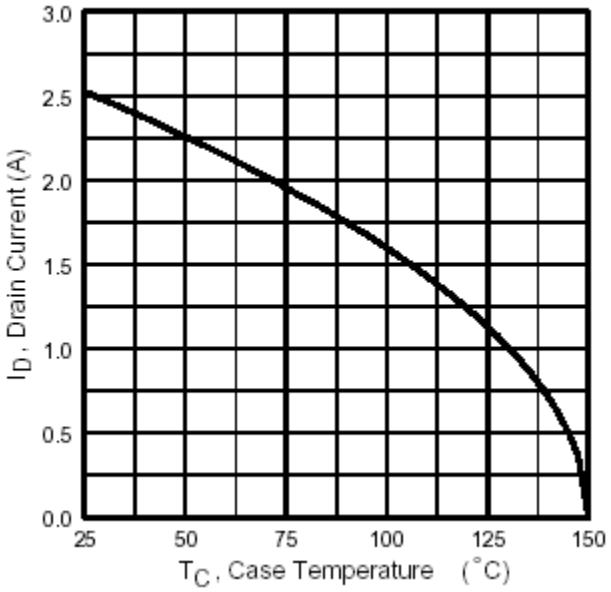


FIGURE 3. Thermal impedance curves.

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2N7463T2, U5



2N7464T2, U5

FIGURE 4. Maximum drain current vs case temperature.

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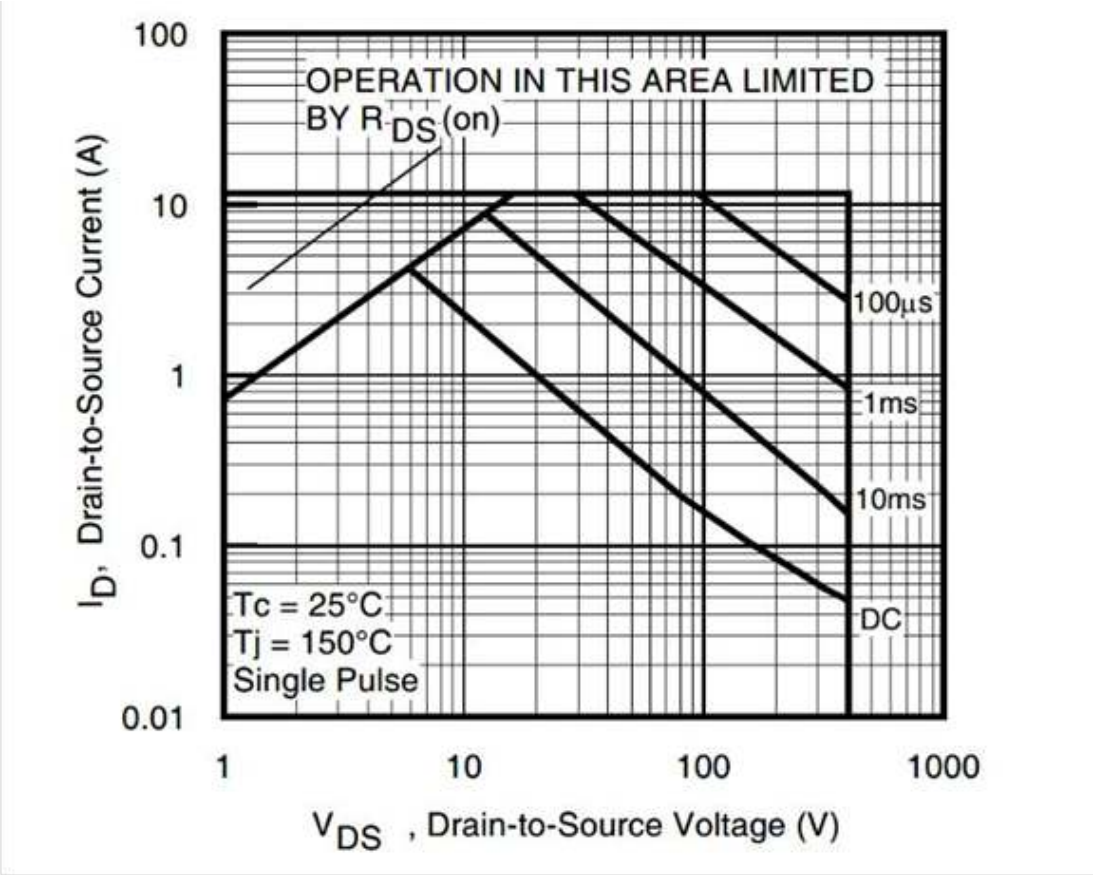


FIGURE 5. Safe operating area graph.

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2N7464T2, 2N7464U5

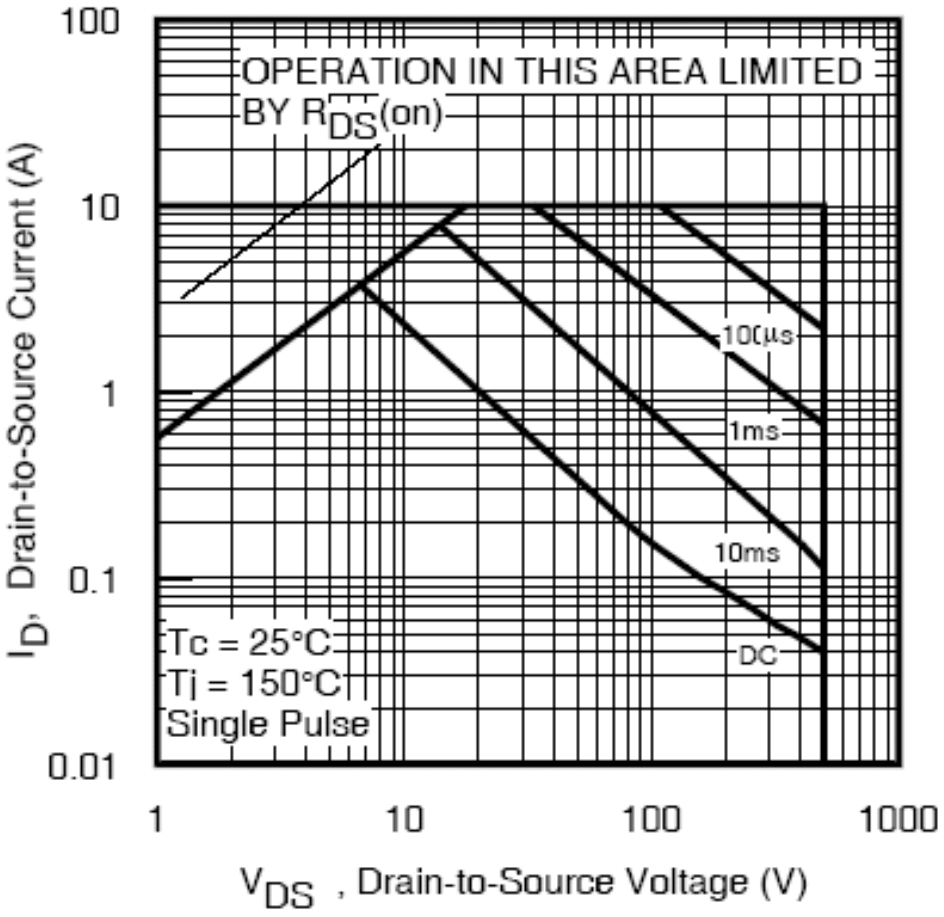


FIGURE 6. Safe operating area graph.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

e. For acquisition of RHA designated devices, table II, subgroup 1 testing of group D herein is optional. If subgroup 1 is desired, it should be specified in the contract or order.

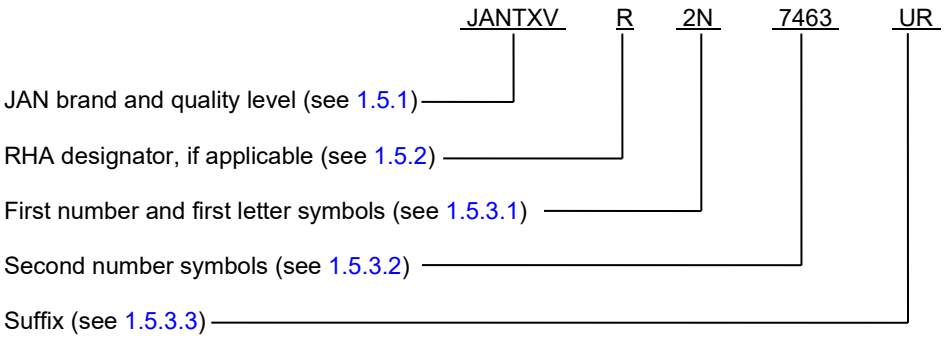
f. If specific SEE characterization conditions are desired (see section 6.5 and table IV), manufacturer's cage code should be specified in the contract or order.

g. If SEE testing data is desired, it should be specified in the contract or order.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

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6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.



6.5 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level
JANTXV#2N7463T2	JANS#2N7463T2
JANTXV#2N7463U5	JANS#2N7463U5
JANTXV#2N7464T2	JANS#2N7464T2
JANTXV#2N7464U5	JANS#2N7464U5

(1) The pound symbol (#) represents the RHA level of “R”.

6.6 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN). This information in no way implies that manufacturer's PIN's are suitable for the military PIN.

Preferred types	Commercial types	
	TO-205AF	LCC
2N7463T2, 2N7463U5 2N7464T2, 2N7464U5	IRHF7330SE IRHF7430SE	IRHE7330SE IRHE7430SE

6.7 Application data.

6.7.1 Manufacturer specific irradiation data. Each manufacturer qualified to this slash sheet has characterized its devices to the requirements of MIL-STD-750 method 1080 and as specified herein. Since each manufacturer’s characterization conditions can be different and can vary by the version of method 1080 qualified to, the MIL-STD-750 method 1080 revision version date and conditions used by each manufacturer for characterization have been listed here (see table IV) for information only. SEE conditions and figures listed in section 6 are current as of the date of this specification sheet, please contact the manufacturer for the most recent conditions.

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TABLE IV. Manufacturers characterization conditions.

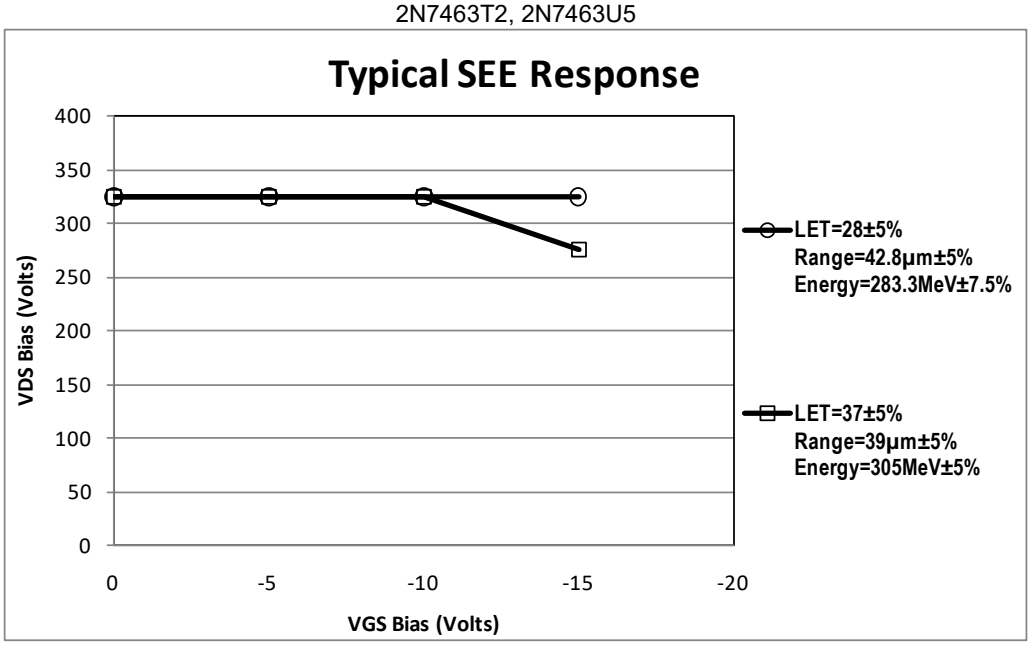
Manufactures cage	Inspection	MIL-STD-750		Sample plan
		Method	Conditions	
69210 (Applicable to devices with a date code of 21 August 2012 and older)	SEE 1/	1080	See MIL-STD-750E method 1080.0 dated 20 November 2006. See figure 7	
	Electrical measurements		I_{GSSF1}, I_{GSSR1}, and I_{DSS1} in accordance with table I, subgroup 2	3 devices
	SEE irradiation:		Fluence = 3E5 ±20 percent ions/cm^2 Flux = 2E3 to 2E4 ions/cm^2/sec, temperature = 25º ±5 °C	
*	2N7463T2 & 2N7463U5		Surface LET = 28 MeV-cm^2/mg ±5.0 %, range = 43 µm ±7.5%, Cu energy = 285 MeV ±7.5% (nominal 3.86 MeV/nucleon at Brookhaven National Lab Accelerator) In-situ bias conditions: V_{DS} = 325 V and V_{GS} = -15 V (typical 4.53 MeV/nucleon at Brookhaven National Lab Accelerator)	
	2N7464T2 & 2N7464U5		In-situ bias conditions: V_{DS} = 375 V and V_{GS} = -20 V (nominal 4.53 MeV/nucleon at Brookhaven National Lab Accelerator)	
*	2N7463T2 & 2N7463U5		Surface LET = 37 MeV-cm^2/mg ± 5%, range = 39 µm ±5%, Br energy = 305 MeV ±5% In situ bias conditions: V_{DS} = 325 V and V_{GS} = -10 V V_{DS} = 275 V and V_{GS} = -15 V (nominal 3.82 MeV/nucleon at Brookhaven National Lab Accelerator)	
	2N7464T2 & 2N7464U5		In-situ bias conditions: V_{DS} = 350 V and V_{GS} = -10 V V_{DS} = 325 V and V_{GS} = -15 V V_{DS} = 300 V and V_{GS} = -20 V (typical 3.82 MeV/nucleon at Brookhaven National Lab Accelerator)	
	Electrical measurements		I_{GSSF1}, I_{GSSR1}, and I_{DSS1} in accordance with table I, subgroup 2	
Upon qualification, all manufacturers shall provide the verification test conditions to be added to this table.				

1/ I_{GSSF1}, I_{GSSR1}, and I_{DSS1} was examined before and following SEE irradiation to determine acceptability for each bias condition. Other test conditions in accordance with table I, subgroup 2, may be performed at the manufacturer's option.

* 2/ Manufacturer performed heavy ion SEE(SEB/SEGR) test at Brookhaven Radiation Effects Facility in accordance with TM1080 of MIL-STD-750. No single event burnout (SEB) and Single event gate rupture(SEGR) were observed to surface LET as stated above table IV and safe operating area (see figure 7). Limits are characterized at initial qualification and after any design or process changes which may affect the SEE(SEB/SEGR) characteristics. For more information on SEE (SEB/SEGR) test results, customers are requested to contact the manufacturer.

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2N7463T2, 2N7463U5



2N7464T2, 2N7464U5

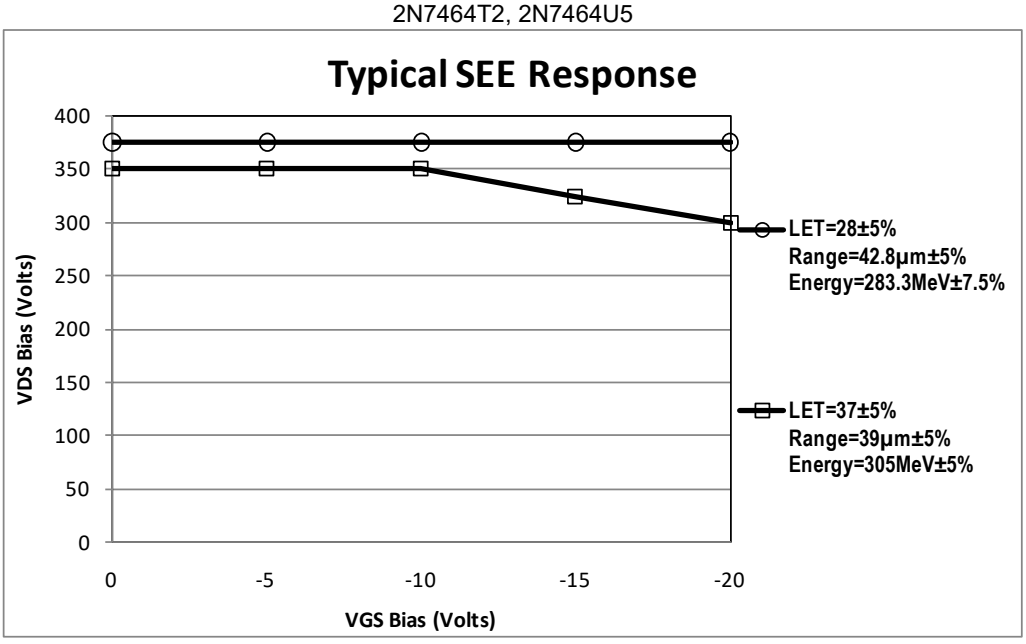


FIGURE 7. Single event effects safe operating area graphs.

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* 6.8 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil.

6.9 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
* Navy - SH
Air Force - 85
NASA - NA
DLA - CC

Review activity:
Air Force - 19

Preparing activity:
DLA - CC

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